New York Lottery Retail Sales and Location Analysis

Yi Fang Kao, Ransom MacLeith, Brianna Sabol

Introduction

Retail site selection is, in many circumstances, more of an art than a science. Common quantitative criteria exist, but the application of those criteria to actual site selection continues to heavily rely on the subjective expertise of business leaders and real estate specialists. Given the recent democratization of machine learning methods, we wanted to explore whether the complex relationship between site-specific data and retail sales can be usefully modeled and thereby reduce the subjectivity and uncertainty of retail site selection.

The first challenge is that we do not have public access to proprietary sales data. As a result, we chose Lottery Daily Retailer Sales published by the New York State Gaming Commission, who by law have access to lottery retailers' lottery sales records. Because lottery sales occur in a commercial environment that notably differs from other retail goods, our project is meant as an exercise in modeling site-specific data. Applying these results to a commercial use case would require significant adaptation to that businesses's or industry's specific goals and relevant market factors. For our project, we examined how a retailer's geospecific environment might relate to lottery sales across New York State. Our goals are to predict expected lottery sales for a theoretical new retail location, and to determine whether lottery retailers fall into distinct archetypes based on their surrounding retail environment.

Supervised machine learning methods will attempt to predict the continuous outcome variable (average daily sales), using features such as population density, median income, proximity to existing retailers, and walkability index. Specifically, we will explore this using Ridge regression, Random Forest, and Gradient Boost models to determine which results in the best fit model. The primary unsupervised methods that we used are Principal Component Analysis (PCA) and Gaussian Mixture Models (GMMs). We tried to use PCA to represent the raw feature set with a reduced number of representative latent dimensions. We used GMMs to investigate whether lottery retailers occupy distinct archetypes of retail environments.

The democratization of machine learning tools can appear to provide solutions wherever a high volume of data is available. However, our results support the hypothesis that most of the variation in sales is driven by factors and decisions specific to the individual retailer. As a result, the persistence of subjectivity in retail site selection is, for the time being, likely appropriate. For machine learning to bring data-driven efficiency, statistical support, or certainty to retail site selection, it would need access to data that sufficiently represents the decisions a retail location makes and the specific circumstances that it adopts. As a result, any improvement in predictive performance is far less about optimizing the data already available and far more about creating a data strategy and infrastructure that supports the systematic measurement of site-specific factors. This kind of effort would likely only be worthwhile for large chain retailers with hundreds or thousands of locations, making lottery sales and independent convenience stores in general an unlikely sector to realize these gains in the near future.

Related Work

Despite large-scale involvement of the lottery market in the economy, there is limited academic research around lottery sales, in comparison to other retail sectors (Cao et al., 2026). However, a few studies were referenced when developing our project proposal. First, Ting et al. (2018) combined population, location, education status, property price, and monthly sales data from 101 telecommunications branches in Malaysia to estimate sales at a new location using metrics such as Euclidean, Jaccard, Manhattan, and Gower distance. Our first goal for this proposal is similar: predicting a retail location's sales from available features. However, we will train and compare predictive models, rather than relying on nearest-neighbor similarity. We will also provide additional context by using PCA and GMM to determine whether retailers form distinct, interpretable archetypes. In addition, Cao et al. (2026) used historical lottery sales data for China from July 2008 to December 2025 to propose a framework that integrates Prophet and SARIMA, for accurately forecasting lottery sales. Our proposal is different in that we use lottery sales data only from one year; however, we use a multi-step process to predict lottery sales, where findings from unsupervised learning inform decisions for supervised learning models.

Our unsupervised efforts to identify environmental archetypes have more directly related academic parallels. The International Journal of Health Geographics attempted to represent the relationship between neighborhood environments and the availability of sit-down restaurants and supermarkets (Peng et al., 2021). To do so, they conducted cluster analysis based on sociodemographic characteristics (e.g. population density, land use mix) and identified six neighborhood types in the St. Paul / Minneapolis region. They then used mixed effects regression models to estimate differences in the availability of sit-down and supermarkets across three decades. Our unsupervised goals and approach are similar with two differences. First, to the extent possible, each lottery retailer's environment is specific to that site, meaning that we are clustering individual retailers, not neighborhoods or census block groups. Second, we do not attempt to compare cluster analyses across time periods. Our resultant environmental archetypes describe retailers solely with the data available in 2025.

Data Sources

	Dataset Description	How to Access	Year	Num Records
NY Lottery Sales	Each record is one lottery retailer's sales by lottery game type for a single day.	Socrata client API query	2025	11M+
Overture Maps	Each record is one place identified by Overture Maps. Places include all buildings and landmarks with a verifiable online presence.	Query Amazon S3 repository	2025	393,891
LODES	RAC: each record is workforce data on the residents of a single census block. WAC: each record is workforce data on the people who work in a census block.	Download csv.gz files	2023	223,405
AADT	Each record is a road segment whose traffic volume is monitored by the NYS Department of Transportation	Download gbd.zip files	2024-2025	49,962
EPA	Each record is walkability and density data for a single census block group.	Download csv file.	2021	15,463
ACS	Each record is data collected through ACS 5-year Community Surveys for a single census block group.	Query the US Census API	2020-2024	16,031

Feature Engineering

1. Geocode all lat-long using the US Census geocoder and retailer addresses to ensure that retailer locations and location-specific features are on the same coordinate reference system. Use the geocoded lat-long to assign a census geoid, which includes census tract and block group
2. Aggregate retailer sales figures. Many retailers were inactive for part or all of the year. To calculate 'avg_daily_sales', use active selling days to distinguish retailers that were only active for part of the year.
3. Overture Maps: use sklearn's BallTree to calculate place counts within a specified radius for each retailer.
4. LODES: convert absolute employment counts to proportions to reduce collinearity with total population
5. AADT: Use geopandas' spatial join to retain the highest-volume road segment with a specified radius.
6. Merge features by census block group (LODES, EPA, ACS) or by lat-long location (Overture, AADT).
7. Drop redundant and highly collinear features. For instance, we dropped household density and count variables for collinearity with population density and count.
8. Conduct a grouped train/val/test split by census tract (not block group) to mitigate spatial data leakage. Because census block groups are relatively small, retailers in adjacent block groups can share highly similar characteristics. Splitting by census tract (one level higher) reinforces our objective of modeling a larger geographic region, as opposed to relying on comparison to proximate retailers.

See Appendix A for the list of all final features.

Initial preprocessing required to handle noisy or missing data

After initial data cleaning, we divided preprocessing between supervised and unsupervised pipelines. Both preprocessing pipelines impute numerical missing values by median, apply np.log1p() log transformations for significantly right-skewed features, and standardize values using StandardScaler(). Because unsupervised methods' objective of describing potential retail environment archetypes differs from the objective of supervised sales prediction, the unsupervised pipeline makes feature selection and engineering decisions that were not replicated in the supervised pipeline. For instance, whereas the supervised pipeline retained all features, the unsupervised pipeline removed categorical features, aggregated some employment-related features into broader categories, and trimmed the feature set to 18 features designed to avoid disproportionate weighting of feature categories. The supervised pipeline also log-transformed three variables that the unsupervised pipeline did not. In supervised sales prediction, reducing the influence of extreme values on standardized distributions can be more important. Contrarily, PCA and clustering methods can learn meaningful environmental variation from large relative differences, even when the underlying distribution has moderate-high skew, encouraging the unsupervised pipeline to apply transformations more conservatively.

Part A (Supervised Learning)

Methods description

After conducting preprocessing, we selected both our candidate model families and our candidate feature presentations. For model families, we selected Ridge regression, K-Nearest-Neighbors, Random Forest, Gradient Boosting to evaluate whether increasingly flexible modeling approaches improved lottery sales prediction. Ridge regression provides an interpretable linear baseline while using L2 regularization to reduce overfitting and mitigate the effects of correlated predictors. Random Forest models may be well-suited to capture the kind of conditional, nonlinear feature interactions we might expect from retailers across different geographic regions, while averaging across trees can help reduce the influence of extreme training observations. Gradient Boosting models work to sequentially correct the errors of preceding trees, enabling it to potentially capture more complex predictive relationships, along they may introduce greater risk of overfitting to the training data and greater dependence on hyperparameter selection. KNN regression, as an instance-based model, tests a different hypothesis that retailers with similar retail environment profiles in multidimensional feature space may tend to have similar sales.

We began with Ridge regression, and after trying multiple combinations of features, including loading PCA and GMM features in addition to or in replacement of the original features, we were unable to find a model that performed well on the training or validation data. At first glance, this indicated our data may be too complex to be captured by linear, additive relationships, so we quickly moved to more sophisticated supervised learning methods. For the random forest, gradient boosting, and KNN regressors, we compared feature representations with group 5-fold cross validation. We then tuned the models on the best representation and evaluated on our validation dataset. The range of tuning settings for each explored hyperparameter are listed below.

Random Forest	n_estimators: [200, 300], max_features: [0.3, 0.5, 0.7, 1.0],	min_samples_leaf: [5, 10, 15, 20, 40], max_depth: [None, 10, 20]
Gradient Boosting	learning_rate: [0.02, 0.05, 0.1], max_iter: [200, 300, 500], l2_regularization: [0, 1, 10]	min_samples_leaf: [20, 40, 60, 80, 150], max_depth: [3, 4, 6, None],
KNN	n_neighbors: [5, 15, 30, 60, 100] distance_metric: [1,2] (Manhattan or Euclidean)	weights: ['uniform', distance']

Supervised evaluation

Overall Results Reporting:

Filepath for full version csv result files: src/supervised/outputs/model outputs

Evaluation Metrics: We used R^2 to describe how much variance in sales each model captures relative to the data's mean average daily sales. For measuring error, MAE in dollars expresses the average difference between predicted and actual daily sales in easily interpretable units. RMSE penalizes large errors more heavily, which helps us assess prediction errors on extreme daily sales values. Spearman correlation helps us evaluate if models can rank retailers as relatively high- or low-performing, even if sales predictions are inaccurate. Applying the same metrics to a validation set of different census tracts evaluates whether relationships learned from the training data can generalize to unseen data. Our models are evaluated on a log scale since our prediction target was log average daily sales. We included evaluation metrics in dollars for interpretability and commercial context.

Model Comparison: The cross validation folds were grouped by census tract. The standard deviations were calculated from the full model results in our ablation study. Random Forest and Gradient Boosting had a very similar performance. Random Forest is our best model based since it performed slightly better on the validation set.

Model	Valudation R^2 (log)	Test R^2 (log)	Cross-validated R^2 (log)
Ridge Regression	0.199	0.229	0.212±0.023
KNN	0.207	0.259	0.240±0.019
Random Forest	0.257	0.290	0.267±0.022
Gradient Boosting	0.254	0.258	0.261±0.022

Comparing Evaluation Metric Between Models: Our results are stable, and the model type did not make a big difference. Random Forest has a dollar RMSE of $2577, nearly twice of the MAE in dollars. Which suggests that a few large errors have a strong effect on the results.

model	**split**	**MAE (log)**	**RMSE (log)**	**R2 (log)**	**Spearman**	**MAE ($)**	**RMSE ($)**	**R2 ($)**	**cv R2 (log)**
random_forest	train	0.483	0.662	0.677	0.871	996.44	2174.45	0.432	
random_forest	val	0.747	0.986	0.257	0.496	1371.62	2576.96	0.139	0.267
gradient_boost	train	0.67	0.875	0.435	0.68	1302.55	2570.03	0.207	
gradient_boost	val	0.751	0.988	0.254	0.488	1379.55	2612.43	0.115	0.261
KNN	train	0.734	0.977	0.295	0.515	1395.37	2617.35	0.178	
KNN	val	0.772	1.019	0.207	0.458	1409.32	2566.35	0.146	0.240

Model Evaluation Visualizations:

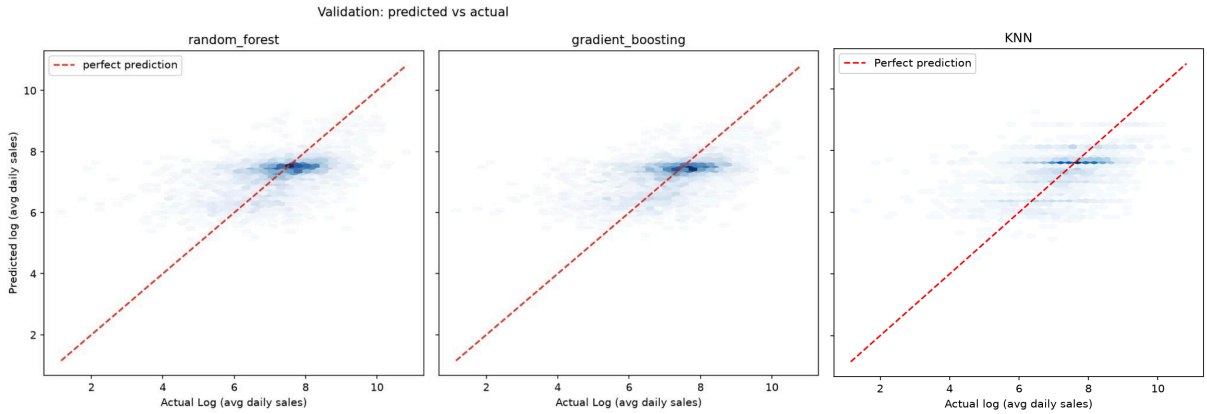


Figure: predicted against actual log average daily sales. Darker means more retailers. The dashed line is a perfect prediction. The figure indicates that extreme sales are the hardest to predict. Most predictions fall between 7 and 8 on the log scale. Retailers with very low actual sales are predicted too high, and retailers with very high actual sales are predicted too low.

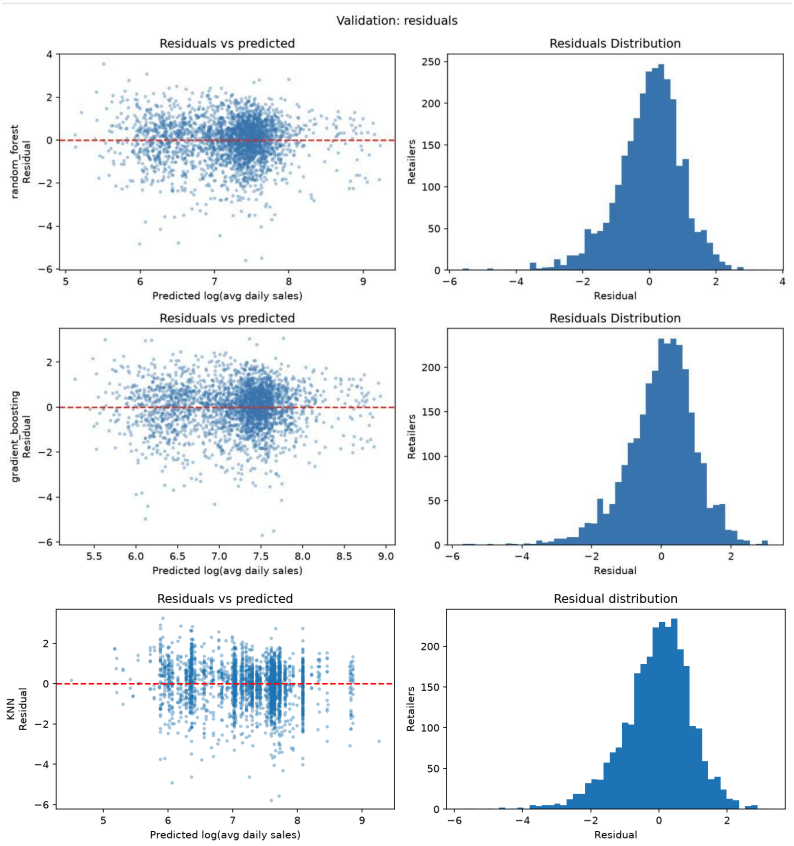


Figure: Left: Residuals against the predicted value. Right: distribution of the residuals. The scatter plot showed that the error spread is fairly consistent. There is no sign that prediction errors get larger as predicted sales increase. The histograms peaks at about 0, indicating that models are equally overpredicting or underpredicting log sales. Random Forest and Gradient Boosting models have similar error patterns, which indicates that neither model has a clear advantage in the types of error it makes. KNN residuals appear vertically striated because the best model used only one-hot encoded categorical variables.

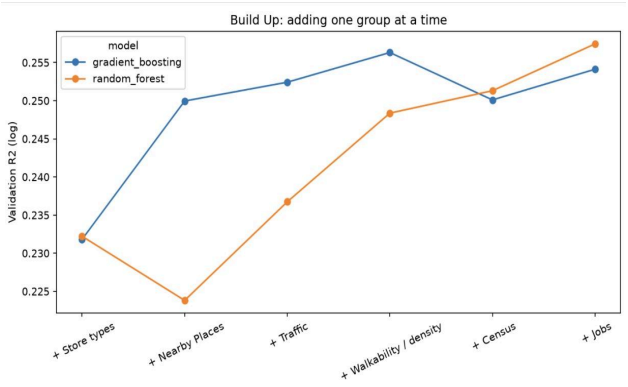
Ablation:

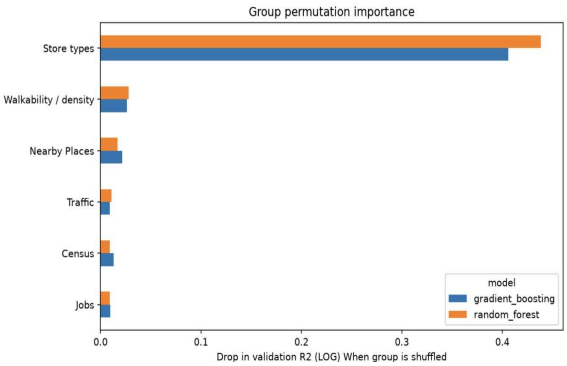
We sorted the model inputs into six groups and tested how each group affected model performance. For each group, we retrained both tuned models with the group removed, using only that specific group, and by adding groups one at a time with the same tuned setting. We also shuffled the values in each group in the validation set to see how much performance dropped.

Result Summary

Group	Feature Counts	R^2 without the group	R^2 with only the group
All Groups	27	0.257	
Store Types	3	0.04	0.232
Traffic	1	0.257	-0.09
Nearby Places	3	0.263	-0.010
Walkability & Density	4	0.267	0.001
Census	10	0.269	0.006
Jobs (LODES)	6	0.265	0.004

The model depends almost entirely on the three store types results (business type, chain, region). Store types alone scored 0.232 for both models which is about 90% of the full model scores. Without store types drops validation R^2 from 0.257 to 0.040, and the mean error increases from $1372 to $1541. Within store types , business type is the most important feature, removing bus type drops the R^2 to 0.081 for Random Forest and 0.077 for Gradient Boosting. The neighborhood groups make little difference. The change in R^2 is 0.01 or less when any one of these groups is removed. Overall, the results suggest that neighborhood features add little new information to the models. Removing them has almost no effect, and using them alone does not produce good predictions.





Sensitivity analysis:

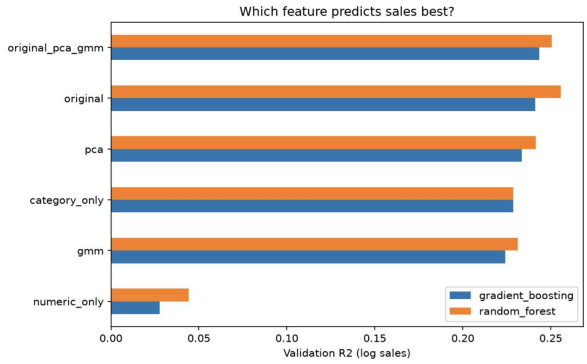
Summary result chart

Input used	Validation R^2	Input used	Validation R^2
All inputs	0.257	Drop traffic	0.257
Store types only	0.232	Drop nearby places	0.252
Everything expect store types	0.040	Drop walkability / density	0.255
Drop business type	0.081	Drop Jobs	0.251
Drop chain	0.240	Drop Census	0.258
Drop region	0.247		

Tuned Setting - Random Forest

Setting	Validation R^2	Cross-validated R^2
Default	0.256	0.264
Tuned	0.257	0.267

We tested how sensitive our best model, the Random Forest, is to the inputs it's given and to its tuning settings. All scores are R^2 on log sales. Cross-validation is grouped by census tract, and its fold to fold spread is about ± 0.022, meaning the differences of 0.02 or less are noise. From our tests, the results depend heavily on the store type input, and less on the others or the tuning. We retrained the Random Forest model with each input removed, and with only the store type input. From the chart, we can visualize that the results are sensitive to business types. For tuning settings, we searched 10 possible Random Forest settings with tract grouped cross validation. The tuning results showed little to no difference. Validation changed by 0.001 and cross-validated by 0.003. We replaced, or added to the original neighborhood columns with the unsupervised features: PCA components and GMM cluster probabilities. The results showed that PCA and GMM recover only part of what the original columns add to the store labels (0.242 and 0.231 against 0.256). Every set that includes the store types results between 0.229 and 0.257 on validation.



Tradeoff: The first tradeoff we experienced was determining between our random forest regressor model and the gradient boost model. The log R2 score for the random forest model was 0.677 for the training dataset and 0.257 for the validation dataset. Conversely, the log R2 score for the gradient boost model was 0.435 for the training dataset and 0.254 for the validation dataset. The log R2 score was similar among the validation datasets for both models; however, the log R2 score was slightly better for the random forest model, which is why we chose this as the best-performing model.

However, when taking a closer look at the random forest model, there are more tradeoffs to note. In addition to the decrease in log R2 score, the log MAE score increases from 0.483 to 0.747, when applying the model to the training data versus validation data. Furthermore, the Spearman correlation score drops from 0.871 on the training data to 0.496 on the validation data. All of which point to overfitting of the model, which demonstrates a tradeoff between learning the training data and generalizing well to unseen data.

Lastly, the random forest parameters that resulted in the best evaluation metrics (n_estimators: 200, min_samples_leaf: 5, max_features: 0.5, max_depth: None) had a mean fit time of 0.72 seconds. In contrast, the fastest model training time was 0.30 seconds; however, it was seventh in predictive performance. The model that was worst in predictive performance had a training time of 1.3 seconds. This points to a tradeoff between computational efficiency, predictive performance, and model generalizability. On one hand, the best-performing model requires twice the amount of training time. In this instance, the performance improvement is justified by the additional training time; however, with larger datasets the additional time could have more consequences. Yet by allowing unlimited tree depth, model complexity increases and introduces the potential for overfitting.

Failure Analysis: Failure analysis was conducted by looking at the largest validation errors from the random forest regression. Specifically, taking the actual minus predicted log values and the actual minus predicted sales values, resulted in categorizing the error direction as either an underprediction or overpredication as well as whether the sales were unusually low or high. Sorting by the largest log-sales errors yielded three retail locations ("agtno"), each in a different error category.

Store location 1 ("agtno" = 112801) had an overprediction of unusually low sales, where the actual sales were $5.15 and predicted sales were $1466.77. A SHAP plot demonstrates the features "aadt_count" (Feature A) reduced the sales prediction, while "bustype_RESTAURANT-HOTEL" (Feature B) and "bustype_DRUG STORE" (Feature C) increased predicted sales. This points to an edge case failure where the low sales may have been poorly represented in the training data, and could be improved upon by ensuring rare low-sales cases are included in the training dataset.

Store location 2 ("agtno" = 112838) had an underprediction of unusually high sales, where the actual sales were $8677.83 and predicted sales were $1582.49. A SHAP plot demonstrates the features "bustype_GAS" (Feature A) and "bustype_RESTAURANT-HOTEL" (Feature B) increased the sales prediction, while "bustype_CONV. STORE" (Feature C) decreased predicted sales. This points to an edge case failure where the high sales may have been poorly represented in the training data, and could be improved upon by ensuring rare high-sales cases are included in the training dataset.

Store location 3 ("agtno" = 115618) had an overprediction of a typical sales range, where the actual sales were $247.26 and predicted sales were $1044.34. A SHAP plot demonstrates the features "bustype_MISCELLANEOUS" (Feature A) and "bustype_CONV. STORE" (Feature B) decreased the sales prediction, while "bustype_RESTAURANT-HOTEL" (Feature C) increased predicted sales. For future improvements, it is worth checking whether similar businesses have the same issue with overprediction, otherwise, this may represent an unusual circumstance.

Overall, this failure analysis reveals the features that had the most impact on underprediction or overprediction of usually high or low sales at all three store locations were different store/business types. This is consistent with the findings from our ablation analysis. The ablation analysis demonstrated the random forest model largely depends on store/business types; thus, it is reasonable they would show up as features with the most impact on errors.

	Feature A		Feature B		Feature C	
Agtno	Transformed value	SHAP value	Transformed value	SHAP value	Transformed value	SHAP value
112801	-1.79	-0.18	0.00	0.11	0.00	0.05
112838	1.00	0.11	0.00	0.097	0.00	-0.08
115618	1.00	-0.14	0.00	-0.11	0.00	0.08

Part B (Unsupervised Learning)

Our unsupervised learning methods aim to answer two primary questions.

1. Can we identify stable and meaningful archetypes of lottery retailers based on their surrounding socioeconomic and built-environment characteristics?
2. Can those archetypes provide useful features for supervised sales prediction?

Feature Representation

Before exploring any specific unsupervised methods, we first had to decide the feature set to use. We divided the raw feature set broadly into six categories:

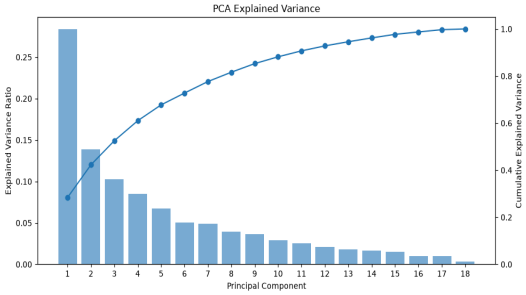
Competitive Environment	Competitors within 1mi
Surrounding Activity	Commercial places within 1mi, Community places within 1mi
Traffic	Traffic volume and number of lanes on nearest measured road segment
Accessibility	EPA walkability index, Road network density
Sociodemographics	Total population, Population density, Median household income, Proportions of age groups, poverty rate, higher education rate, and proportions of households that own their home, live with their family, or own a vehicle.
Employment / Workforce	Labor force participation rate, Number of jobs held by residents, Number of jobs in the area, Proportion of jobs that are low- and medium-earning, Proportion of jobs in various employment sectors

The raw feature set is relatively dominated by sociodemographic and employment variables. Including a disproportionately large number of overly correlated or conceptually similar variables describing one category, such as employment, can bias the clustering process at the expense of other feature categories. To engineer a more balanced feature representation, we first addressed the proportion of jobs in various employment sectors. LODES WAC data distinguishes between 20 different employment sectors. We consolidated those sectors in four general categories: consumer facing jobs, office jobs, industrial and logistics jobs, and supporting services (e.g. gov't). We then dropped features for high correlation and/or conceptual redundancy. For instance, of the residence area and workplace area job counts and proportions, we kept only the total jobs held by residents and the proportion of low-earning jobs, as including both rac and wac data would bias the data in favor of employment earnings tiers. Overall, we trimmed the feature set from 51 original features to 18 used in unsupervised learning.

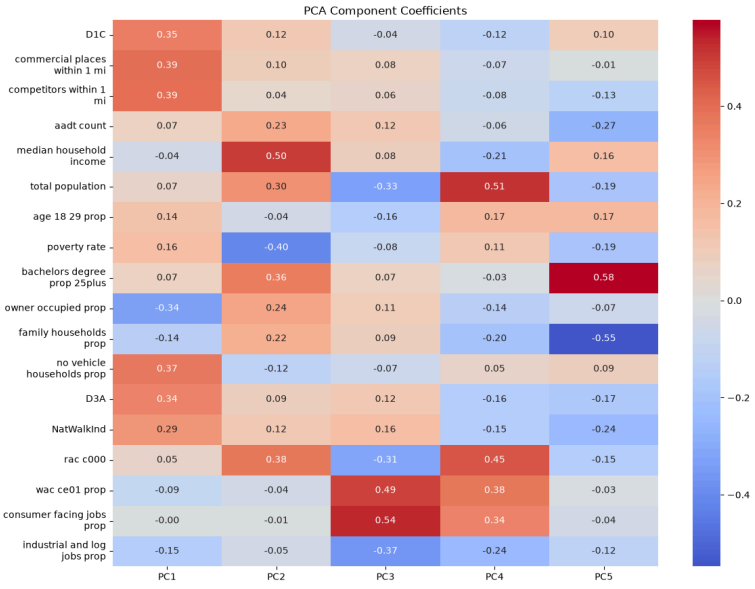
Principal Component Analysis

After deciding on a reasonably balanced feature representation, we applied Principal Component Analysis (PCA).. First, we sought to use PCA to reduce dimensionality and uncover potential latent dimensions of a retailer's environment. As part of data cleaning, we removed or selected between features with extremely high correlations. However, many of the retained features nevertheless had moderately high correlations, potentially reflecting common underlying characteristics. For instance, we noticed absolute correlations between 0.72 and 0.81 for competitors within 1mi, proportion of households without a vehicle, proportion of households occupied by the owner, community places within one mile, and population density. A reasonable hypothesis is that these features and potentially others collectively reflect the degree of urbanization surrounding a retailer. PCA transforms the original features into a smaller set of orthogonal components (for instance, an 'urbanization' component) that aim to preserve information captured by the original features while reducing redundancy and correlation.

Secondly, we hoped that the resultant feature representation as a reduced set of principal components could be used for both downstream unsupervised clustering and supervised sales prediction. The PC feature sets did significantly improve clustering performance but ultimately failed to meaningfully support supervised sales prediction. Ultimately, we produced the following PCA results:



PC1: 28.4% explained variance
PC2: 13.9%
PC3: 10.2%
PC4: 8.5%
PC5: 6.7%
5 PCs explain 67% of the variance, 7 PCs explain 78%, and 11 PCs explain 91%.



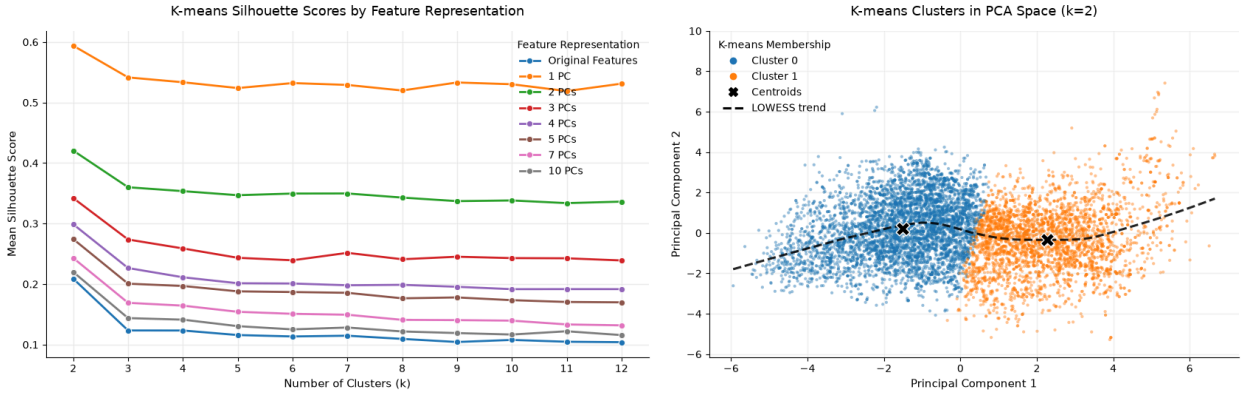
PC1 combines correlations in employment density, place counts, road network density, not having a vehicle, and home ownership into a latent dimension that reflects **"Urbanization / Commercial Density"**.
PC2 combines correlations between median household income, poverty rate, employment counts, population count, and educational attainment into a latent dimension that reflects **"Socioeconomic Advantage"**.
PC3 distinguishes between environments with more **"Consumer-Facing, Low-Earning Jobs"** and environments with more industrial jobs and residential land use.
PC4 and PC5 are relatively less interpretable. PC4 captures variance by population scale and employment composition not already captured by PC1 and PC3. PC5 appears to distinguish between different kinds of suburbs by education and household composition.

Unsupervised Clustering Evaluation Metrics

- **Silhouette Score** measures cluster quality using cohesion (average distance between each point to all other points in the same cluster) and separation (average distance between each point and all points in the nearest neighboring cluster).
- **Bayesian Information Criterion (BIC)**, specifically in the context of evaluating a Gaussian Mixture Model (GMM), attempts to maximize the log-likelihood of the observed data while penalizing the use of too many Gaussian components to do so. After fitting the GMM, we also measured the log-likelihood of the validation set.
- **Adjusted Rand Index** across repeated model fits was used to evaluate cluster stability by measuring the consistency of cluster assignments across different initializations.

Together, these metrics provided a picture of the quality, generalizability, and stability of our cluster assignments.

K-Means Baseline



There are several insights that in conjunction support one central argument about lottery retailer environments. First, using principal components does improve k-means clustering relative to the original features. Second, using more PCs only detracts from the silhouette score; in fact, using just the first PC representing urbanization/commercial density yields the highest silhouette score of nearly 0.6. Third, k-means consistently performed the best when dividing retailer environments into just two clusters. Fourth, the scatterplot of clusters in PC1-PC2 feature space resembles a continuous distribution, as opposed to having identifiably separable clusters.

Without separable clusters of retailers, the k-means algorithm takes advantage of the strongest separator that it has, a retailer's position along PC1, which captures a sizable 28% of the variance in retailer environments. These results suggest that retailer environments vary along a dominant continuous gradient from less urbanized to more urbanized but do not form clearly separable groups. Additional principal components capture differences in sociodemographic and employment characteristics that are descriptively meaningful but do not improve cluster cohesion and separability using a k-means algorithm. As a result, the addition of more PCs primarily serves to increase within-cluster dispersion relative to between-cluster separation, reducing silhouette scores.

These results also help focus our subsequent clustering efforts. We originally considered HDBSCAN for its ability to identify clusters with varying densities, but the lack of naturally separable clusters entirely makes this an unlikely solution. A quick experimentation with HDBSCAN confirmed this intuition. Across multiple hyperparameter configurations and feature representations, HDSCAN consistently produced one of three outcomes: (1) one dominant cluster containing more than 90% of retailers, (2) classifying most (>75%) of retailers as noise, and (3) with relaxed parameters, identifying hundreds of small clusters with low persistence. Both the k-means and HDBSCAN strengthen the case for exploring Gaussian Mixture Models, as probabilistic cluster membership and flexible covariance structure may enable it to better represent the data as a set of complex, overlapping distributions.

Gaussian Mixture Models (GMMs)

Unlike k-means, which divides retailers into mutually exclusive, globular clusters, GMMs represent the data as a weighted mixture of Gaussian components. Each Gaussian component is described by its mean (position), covariance matrix (shape and orientation), and weight (size). Critically, GMMs with a 'full' covariance type allow each component to have its own covariance matrix, enabling clusters to be ellipsoids of any shape and direction, more appropriate for feature environments that vary along continuous, overlapping dimensions. In addition, whereas k-means makes hard cluster assignments, GMMs assign assignment probabilities to each retailer. This more nuanced representation of a retailer's environment is more likely to be useful for downstream supervised prediction tasks.

We started by identifying competing feature representations, including the original standardized features as well as a range of principal components. For each feature representation, we conducted a grid search across the number of Gaussian components and covariance structures. Within each experiment, models were primarily evaluated using the Bayesian Information Criterion (BIC) and supplementarily evaluated using log-likelihood of the validation data set. We did not compare BIC and log-likelihood across feature representations, as they are not directly comparable.

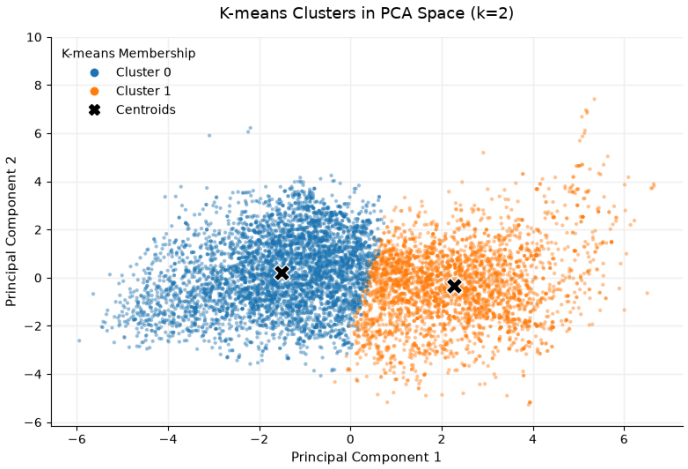
We identified two foundational insights. First, the best-performing GMMs all utilized full covariance matrices, allowing for maximally flexible component shapes and directions. Second, no GMM fit on more than five PCs was able to arrive at a stable result. We believe additional dimensions increased model complexity without improving distinct clustering structure sufficient to support reproducible component assignments. We also identified two candidate GMMs: an eight-component GMM fit on three PCs, and a five-component GMM fit on five PCs. The two candidates were evaluated on several criteria. Critically, the GMM most helpful for identifying retail environment archetypes and the GMM most helpful for supervised sales prediction do not have to be the same model.

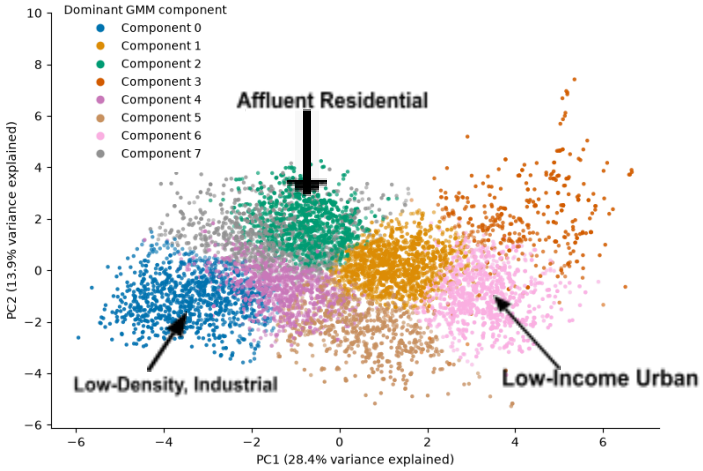
Criterion	**3 PCs, 8 Gaussian components**	**5 PCs, 5 Gaussian components**
Variance Retained	52.5% of original feature set variance.	**67.7% of original feature set variance.**
Initialization Sensitivity (25 random seeds, n_init=30)	**Clearly superior. 0.978 mean pairwise ARI over 300 pairwise comparisons.**	Moderate. The GMM arrives at two different locally optimal solution families, dependent on random seed. ARI within each solution family was high. We selected a representative solution from the lower-BIC solution family.
Validation Set Confidence	Mean membership confidence: 69.0%. 36.7% have confidence less than 60%.	Mean membership confidence: 75.7%. 23.6% have confidence less than 60%.

Cluster interpretability	**Less interpretable, more detailed.** Additional components allow for greater differentiation of environments by employment composition.	Broader components enable **more confidently interpretable** components.
Main Advantage	Reproducibility and more detailed environment profiles.	Captures more info from the original feature set. More confidently interpretable clusters.
Main Disadvantage	Greater component overlap. Omits info captured by additional PCs.	Less initialization stability. Fewer clusters may obscure more specific retailer distinctions.

Ultimately, we selected the three-PC, eight-component GMM for interpreting retailer environment archetypes because of its higher initialization stability and ability to distinguish additional subpopulations by socioeconomic and employment characteristics. We selected the five-PC, five-component GMM for supervised feature engineering because it preserves a greater proportion of the original feature set variation while producing a more compact and interpretable set of clusters. We made these selections based on the distinct objectives of the two tasks. Unsupervised interpretation prioritizes stable, detailed environmental profiles, whereas supervised prediction prioritizes the use of features that are interpretable and retain more information from PCA decomposition.

Overall Results Reporting



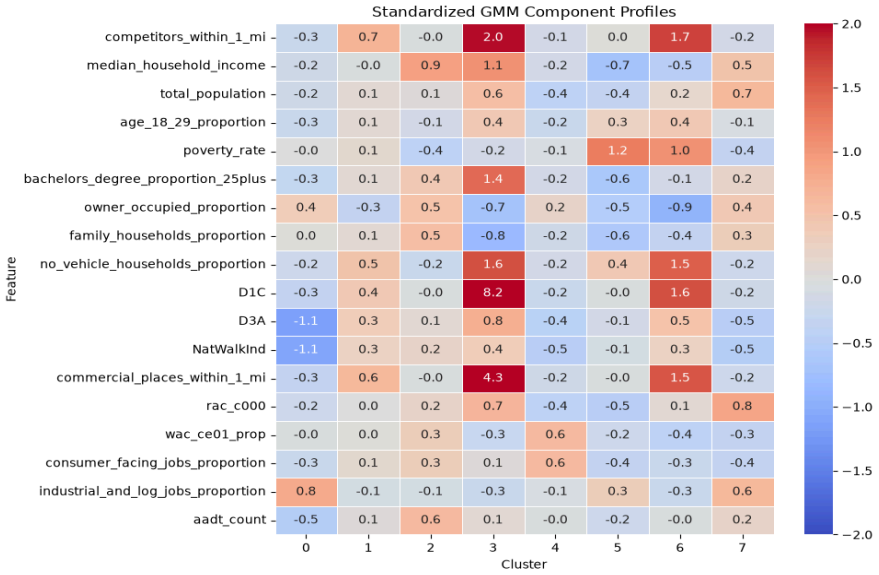


	3 PCs, 2 K-Means Clusters	**3 PCs, 8 Gaussian Components**
Clustering Method	Hard assignments. Spherical covariance. Globular clusters.	**Probabilistic assignments. Flexible covariance. Clusters of varying size, shape, and direction.**
Evaluation Metric	Silhouette Score	BIC, validation log-likelihood
Cluster Interpretation	Simplistic: more urbanized vs. less urbanized	**More complex: distinguishes retailer environments by urbanization, socioeconomics, and employment composition.**
Main Advantage	Consistent two-cluster solution that is easy to interpret	- Flexible cluster shapes provide a more interesting description of the feature space. - Probabilistic assignments better represent the uncertainty of cluster compositions.
Main Disadvantage	Little evidence of well-separated groups. Clusters are primarily a bisection on urbanization.	Components likely reflect overlapping density structures in feature space and cannot be confidently interpreted as natural categories.

K-Means silhouette scores and GMM BIC scores are not numerically comparable, and evaluation metrics were used exclusively to compare models within each model family. We ultimately selected the GMM model for in-depth exploration not for any particular BIC score but rather because it was more adaptable to the continuous feature space and therefore able to uncover a more descriptive representation of retail environments.

Together, our k-means and GMM analyses support two primary conclusions. First, retailer environments exhibit continuous variation rather than clearly separable categories. In our k-means analysis, two clusters divided primarily by urbanization consistently outperformed larger numbers of clusters by silhouette score. Additionally, the standardized PC1-PC2 scatterplot illustrates a zero-centered, continuous distribution. Because our goal was to characterize retailer environments (and not the retailers themselves), it is reasonable that sociodemographic metrics like employment density or poverty rate would vary over a continuous range.

Second, a GMM's covariance flexibility and probabilistic membership assignment enable useful descriptions of complicated multidimensional distributions without requiring separable clusters.



0: Low-Density, Industrial
1: Moderately Urban, Mixed Use
2: Affluent Residential, Consumer-Oriented
3: Highly Urban, Affluent
4: Low-Density, Consumer-Oriented
5: Low-Density, Low Income
6: Highly Urban, Low Income
7: Affluent Residential, Industrial

* See Appendix B for the profile heatmap of the 5-cluster GMM built on 5 PCs, whose GMM probabilities were passed as potential features to supervised prediction pipelines.

The eight Gaussian components overlap considerably, so these profiles should not be interpreted as discrete and naturally occurring archetypes of retail environments. However, the GMM can recognize patterns k-means clustering struggles to identify. The first 3 PCs roughly represent "Urbanization", "Socioeconomic Advantage", and "Consumer-Facing Employment". Accordingly, the GMM can distinguish between dense, urban, affluent environments (Cluster 3) and dense, urban, low-income environments (Cluster 6). This distinction can be hugely important, especially when analyzing consumer purchasing behavior. On the contrary, k-means clustering considers both of these environments as simply in the more urbanized cluster.

Beyond identifying meaningful retailer-environment archetypes, the interpretable feature profiles of both PCA scores and GMM membership probabilities suggested at the time they could have potential use as engineered features for supervised sales prediction. Separately or together, these representations could help supervised models identify relationships between retailer environments and lottery sales that are difficult to capture using the original features alone. In hindsight, we know that PC scores and GMM probabilities did not significantly improve supervised sales prediction. However, we believe this had little to do with our chosen methods and is more likely the result of an original feature set that fails to sufficiently capture the signals needed to predict lottery sales.

Discussion

The most surprising part from our supervised result was that business type mattered much more than we expected. We initially thought neighborhood features, such as census data, traffic, walkability, and nearby places would play a major role in predicting sales. Instead, the business type explained most of the predictive signal. Our main challenge was the preparation of the input features and preventing leakage between stores in the same tract. To clean and prepare our input features, we used consistent preprocessing pipelines to fill missing values, log transform skewed counts, and scale numerical features. Since stores in the same tract share neighborhood features, we avoided data leakage by grouping the training, validation, and test splits by census tract. We also used tract based grouping for every cross-validation fold.

To extend our solution, we recommend selecting a prediction target that is more realistically dependent on geospecific factors. Many store owners focus on the sale of goods or gas at the pump and treat lottery sales as a secondary or passive source of revenue. Two stores in the same place can differ widely in sales simply because only one actively promotes lottery sales. That motivation and related site-specific decisions are not captured in our data, and it possibly explains why sales vary so much between nearby stores, and why our models explain only about a quarter of sales variance. A better target would be one where owners' effort and motivation are a more reliable assumption. We can also improve our solution by selecting features more intentionally based on what drives a store's success. As opposed to starting with the data available to us, we could alternatively start by identifying what makes a retail location successful. We would then check which factor is covered in the data and collect the ones that are missing. This can help us build a more relevant feature set and fairer way to evaluate whether neighborhood characteristics can improve sales predictions

The most surprising and important lesson from our unsupervised results was the difference between separability and describability. Our principal challenge in unsupervised clustering was that retailer environments did not naturally form separable groups. After realizing this through K-Means, we feared GMMs, even after allowing for full covariance structures, would encounter the same barrier. To some extent, it did; GMMs that used more than five principal components consistently did not arrive at a stable solution. However, when using between two and five principal components, GMMs not only found a solution but produced components with interpretable environmental profiles. This limited success despite the clear lack of easily separable clusters helped illustrate the difference between separability and describability, and to understand a potential alternative use case for GMMs. Even when the data does not reliably represent naturally occurring groups in feature space, they can still describe a retailer environment by the full set of probabilities that a retailer might belong to each of the components. Of course, our ability to find descriptive components did not ultimately improve supervised sales prediction, so we were not able to demonstrate this potential utility. However, it does make us curious about GMMs' ability in future projects to find potential structure in what otherwise may appear to be a continuous mass of multidimensional data.

There are three possible extensions to our solution, in order from least to greatest complexity and involvement. First, it would be worthwhile to investigate GMM stability more robustly. Currently, we conduct sensitivity analysis by random initialization alone, with just a single comparison against the validation set. It would be more informative and defensible to evaluate GMM component stability under different subsets of features and across different subsamples of the training data. Second, we could consider building geographically adaptive pipelines. For New York in particular, New York City has an abnormally high volume of retailers and extreme feature values. Perhaps separating NYC from the rest of the state would enable unsupervised methods to discover finer-grained environmental distinctions. Third, we could investigate whether UMAP dimensionality reduction supports supervised sales prediction better than PCA. However, even if it did help predict sales, there is a clear tradeoff in the reduced interpretability of UMAP-produced features.

Ethical Considerations

Supervised Ethical Issue

Since we are combining data from six different sources to create the master dataset, data aggregation concerns are particularly prevalent. Data aggregation refers to the practice of collecting data from disparate sources and combining it together. It serves as an ethical risk because it can potentially lead to identification of people who fall under certain criterion (MacFarland, 2012). This issue is addressed by providing code as well as documentation of and links to each of the data sources; therefore, future users or readers can replicate the same process for creating our master dataset. In addition, our data does not contain information on single persons, it only goes down to the block group level, making it more difficult to identify an individual in the dataset.

Another ethical issue that's present while providing a solution to Part A is known as the matrix of vulnerabilities. This concept speaks to the privacy harms and increased vulnerability poor communities face in the "big data" era. In particular, "low status users may be both unfairly excluded from opportunities (such as access to credit) and unfairly targeted (for example, by predatory marketing strategies) based on determinations made by predictive analytics and scoring systems…" (Madden et al., 2017, p. 56). In particular, for this project, the risk arises when or if the selected new location for lottery retail, based on potential sales data, falls under an area classified as a socioeconomically disadvantaged community. If this is the case, once a location is determined, additional evaluation safeguards, including community input, could be implemented to determine if it's an ethical and beneficial decision for the area.

Unsupervised Ethical Issue

An ethical risk especially prevalent in our unsupervised learning methods is proxy discrimination. In "How our data encodes systemic racism", Deborah Raji remarks on an alarming practice in US police precincts, in which automated systems trained on data from racially biased encounters are deployed to help police officers make policing decisions (Raji, 2020). While the data presumably does not predict on race directly, proxy variables nevertheless may encode information correlated with it, encouraging the perpetuation of existing biases. In a similar vein, our feature set includes neighborhood-level characteristics such as household income, poverty rates, educational attainment, and vehicle ownership, all of which may correlate with race, ethnicity, or other protected characteristics. PCA combines these characteristics into latent variables, while GMM further translates them into probabilistic cluster memberships. Although these transformations are useful for identifying patterns in retailer environments, they introduce additional layers of abstraction that can make it more difficult to recognize when a model's outputs disproportionately emphasize particular demographic groups. The potential consequences of this abstraction depend on how clustering results are used downstream. In theory, using these clustering results to inform commercial decisions could systematically target specific races or ethnicities, even without explicitly considering them as a factor. Deployment in any form is not recommended, but in theory, responsible deployment would require evaluating cluster assignments and decision outcomes across race, ethnicity, and other demographic groups.

Statement of Work

Overall, we met as a group about twice per week, throughout the course of the project, to give updates on progress and discuss the next steps. In the interim, we were in communication over slack. Specifically, Ransom was responsible for project set up and management of our github repository. While Yvonne and Brianna explored external datasets for use with the project, Ransom processed the main NY Lottery Sales dataset and established the merge column (bg_geoid), based on latitude and longitude coordinates of lottery retailers. Once the merge columns were established, Yvonne and Brianna worked on code to merge our respective external datasets to the main dataset. From there, we collectively decided on features to include in the main dataset. From there, we split up programming of the supervised and unsupervised learning sections. Ransom worked on the unsupervised learning, while Yvonne and Brianna worked on exploratory data analysis, preprocessing, and explored supervised learning models. After obtaining results from our analysis, we collectively worked on the project report, mainly splitting up writing by the separate programming sections we were each responsible for.

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Appendix A - Data Dictionary (Cleaned and Ready for Modeling)

Variable	Source Dataset	Data Type	Description	Processing
agtno	NY Lottery Sales	Unique ID	New York lottery unique agent number as identified by the NYS Gaming Commission.	
avg_daily_sales	NY Lottery Sales	Numeric	Average daily sales ($) in lottery tickets	Sum across ticket types and days, divide total by each retailer's number of active days.
chain	NY Lottery Sales	Categorical (binary)	Lottery store is either 'independent' or 'chain'	Numbers greater than 0 in the raw lottery dataset represented a chain id.
bustype	NY Lottery Sales	Categorical	Label for store type (e.g. conv. store, liquor)	
longitude	NY Lottery Sales	NA	Longitude coordinates	Encode using US Census Geocoder
latitude	NY Lottery Sales	NA	Latitude coordinates	Encode using US Census Geocoder
bg_geoid	NY Lottery Sales	NA	Census block group	Assign by spatially joining lat-long coordinates with NYS block group shapefile
tract_geoid	NY Lottery Sales	NA	Census GEOID, which includes census tract and block group	Assign by spatially joining lat-long coordinates with NYS block group shapefile
comp_within_1_mi	NY Lottery Sales	Numeric	Number of lottery retailers within 1 mi	Sklearn BallTree
aadt_count	NY AADT	Numeric	Annual Average Daily Traffic (AADT) count on the closet match road	Spatial Join
aadt_functional_class	NY AADT	Categorical	Road type for the nearest matched road	
aadt_lanes	NY AADT	Categorical	Number of lanes on the nearest matched road segment	Regex to translate str formatting to num lanes
dist_to_read_m	NY AADT	Numeric	Distance (meter) from the retailer to the nearest matched road	Spatial Join
median_household_income	ACS	Numeric	Median household income for block group	
total_population	ACS	Numeric	Total number of people 18+ in block group	Summed age groups

age_18_29_proportion	ACS	Numeric	Proportion of people aged 18-29 in block group	Divided count by total population
age_30_64_proportion	ACS	Numeric	Proportion of people aged 30-64 in block group	Divided count by total population
poverty_rate	ACS	Numeric	Proportion of people below the poverty line among the poverty universe	Divided below_poverty by poverty_universe
bachelors_degree_proportion_25plus	ACS	Numeric	Proportion of people with a bachelor's degree (only) among adults aged 25+	Divided count by adults aged 25+
post_grad_degree_proportion_25plus	ACS	Numeric	Proportion of people with a master's degree or equivalent/higher among adults aged 25+	
labor_force_participation_rate	ACS	Numeric	Labor force participation rate	Divided in_labor_force by population 16+
unemployment_rate	ACS	Numeric	Unemployment rate	Divided unemployed by labor force
owner_occupied_proportion	ACS	Numeric	Proportion of housing units occupied by the owner in block group	Divided count by occupied housing units
family_households_proportion	ACS	Numeric	Proportion of households comprised of a family unit	Divided count by total households
no_vehicle_households_proportion	ACS	Numeric	Proportion of households without a vehicle	Divided count by occupied households
D1B	EPA	Numeric	Gross population density (people/acre) on unprotected land	
D1C	EPA	Numeric	Gross employment density (jobs/acre) on unprotected land	
D3A	EPA	Numeric	Total road network density	
NatWalkInd	EPA	Numeric	Walkability Index	
commerical_places_within_1_mi	Overture Maps	Numeric	Total number of Overture-identified commercial establishments within 1 mile of the retailer.	Sklearn's BallTree. Aggregated place categories: food and drink, entertainment, shopping, lifestyle, lodging
community_places_within_1_mi	Overture Maps	Numeric	Total number of Overture-identified community-oriented establishments within 1 mile of the retailer.	Sklearn's BallTree. Aggregated place categories: community and gov't, healthcare, education, cultural and historic

rac_c000	LODES	Numeric	Total number of jobs held by residents in retailer's block group	
rac_ce01_prop	LODES	Numeric	Proportion of low-earning jobs held by residents in retailer's block group	Rac_ce01 divided by rac_c000
rac_ce02_prop	LODES	Numeric	Proportion of mid-earning jobs held by residents in retailer's block group	Rac_ce02 divided by rac_c000
wac_cns01	LODES	Numeric	Number of agriculture, forestry, fishing and hunting jobs in retailer’s block group	
wac_cns02	LODES	Numeric	Number of mining, quarrying, and oil and gas extraction jobs in retailer’s block group	
wac_cns03	LODES	Numeric	Number of utility jobs in retailer’s block group	
wac_cns04	LODES	Numeric	Number of construction jobs in retailer’s block group	
wac_cns05	LODES	Numeric	Number of manufacturing jobs in retailer’s block group	
wac_cns06	LODES	Numeric	Number of wholesale trade jobs in retailer’s block group	
wac_cns07	LODES	Numeric	Number of retail trade jobs in retailer’s block group	
wac_cns08	LODES	Numeric	Number of transportation and warehousing jobs in retailer’s block group	
wac_cns09	LODES	Numeric	Number of information jobs in retailer’s block group	
wac_cns10	LODES	Numeric	Number of finance and insurance jobs in retailer’s block group	
wac_cns11	LODES	Numeric	Number of real estate, rental, and leasing jobs in retailer’s block group	
wac_cns12	LODES	Numeric	Number of professional, scientific, and technical services jobs in retailer’s block group	
wac_cns13	LODES	Numeric	Number of company and enterprise management jobs in retailer’s block group	
wac_cns14	LODES	Numeric	Number of administrative, support, waste management, and remediation service jobs in retailer’s block group	

wac_cns15	LODES	Numeric	Number of education service jobs in retailer's block group	
wac_cns16	LODES	Numeric	Number of health care and social assistance jobs in retailer's block group	
wac_cns17	LODES	Numeric	Number of art, entertainment, and recreation jobs in retailer's block group	
wac_cns18	LODES	Numeric	Number of accommodation and food service jobs located in retailer's block group	
wac_cns19	LODES	Numeric	Number of other service (except public administration) jobs in retailer's block group	
wac_cns20	LODES	Numeric	Number of public administration jobs in retailer's block group	
wac_c000	LODES	Numeric	Total number of jobs in retailer's block group	
wac_ce01_prop	LODES	Numeric	Number of low-earning jobs in retailer's block group	wac_ce01 divided by wac_ce000
wac_ce02_prop	LODES	Numeric	Number of mid-earning jobs in retailer's block group	wac_ce02 divided by wac_ce000

Appendix B - 5-PC, 5 Cluster GMM Profile

The associated GMM probabilities for each retailer were passed as features into supervised prediction pipelines. They did not ultimately improve model performance.

